

Spatial Neighbourhood Analysis of mIF Data

Student Lab Exercises — Sessions 3 & 4

Systems Biology Lab, University of Warsaw, MIM

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Background

Last week we divided slides into square patches, summarised each patch as a cell-type histogram, and clustered those histograms. This week we take a complementary approach: instead of analysing fixed regions of space, we analyse the **local neighbourhood of every individual cell**.

The result (a *neighbourhood composition vector* or “**bag of cells**”) describes what surrounds each cell. Clustering these bags defines **spatial niches**: recurring cellular microenvironments.

The two questions we are working towards today:

Are the same spatial niches found across all 15 patients in the cohort, and which niches tend to sit next to each other?

Data reminder. Each `_with_celltype.tsv` file has ~1.2 million cells with columns `nucleus.x`, `nucleus.y` (coordinates in **micrometres**), `cell.type`, `tissue.type`, and ROI flags. Cell types: CD15-Tumor, CD15+Tumor, Tcell, Bcell, BnTcell, Macrophage, DC, Neutrophil, other. Typical cell diameter ~12 um, so 50 um is about 4 cell diameters.

A word on BnTcell. This is not a real cell type; it arises from overlapping B and T cell signals in the panel, a known limitation of multiplexed imaging. Treat it as a marker of signal uncertainty rather than a genuine population, and keep that in mind when you interpret any result where it appears prominently.

Plotting tip. With 1.2 million points, always subsample before plotting:
`df.sample(50_000, random_state=42)`

Session 1: Building Neighbourhood Graphs and Composition Bags (90 min)

Goal. A matrix of shape (*N cells x 9*) where each row encodes the cell-type composition of one cell’s local neighbourhood, the raw material for all niche analysis that follows.

Exercise 1.1: Load One Sample and Explore Its Spatial Structure (20 min) (*Skills Refresher*)

This exercise revisits what you did in the previous lab sessions. Use it to warm up, double-check your workflow, and make sure you have a working environment before moving to new material. If everything runs smoothly, you should finish well within 20 minutes.

Take your time here. Rushing past exploration almost always costs you more time later, and this is data worth looking at carefully.

Tasks

1. Load one `_with_celltype.tsv` file. Print its shape and column names.
2. How many cells of each type are present? What fraction of cells are inside the tumour ROI (`in.ROI.tumor_tissue == TRUE`)?
3. Make two side-by-side scatter plots of cell positions (`nucleus.x` vs `nucleus.y`), one coloured by `tissue.type` and one coloured by `cell.type`. Define a fixed colour per cell type at the start; reuse this palette throughout both sessions.
4. Inspect the plots visually:
 - Do T cells appear to cluster spatially, or are they scattered uniformly?
 - Where do tumour cells concentrate relative to the tissue boundary?
 - Can you identify regions that look like immune aggregates?

Hint (relevant functions): `pd.read_csv`, `df.value_counts()`, `df.sample()`, `plt.scatter`, `df.groupby()`

Think about. If you randomly shuffled the cell-type labels, what would the scatter plot look like? How does the real data differ, and what does that imply biologically?

Exercise 1.2: Build a Spatial Neighbourhood Graph (25 min)

Tasks

1. Build a **k-NN graph** ($k = 10$): for each cell, find its 10 nearest neighbours by Euclidean distance in the (x, y) plane. *Watch out:* the query function returns the cell itself as its own nearest neighbour; exclude it.
2. Build a **radius graph** ($r = 50$ μm): for each cell, find all cells within 50 μm . Because this is slow on 1.2 M cells, first work on a random subsample of 20,000 cells.
3. Plot the **distribution of neighbour counts** for the radius graph. Is it unimodal, or do you see multiple peaks?
4. Visualise the neighbourhood of a single cell: plot the focal cell and all its neighbours coloured by cell type. Choose a cell near the tumour-stroma boundary.

Hint (relevant functions): `scipy.spatial.cKDTree`, `tree.query(coords, k)`, `tree.query_ball_point(coords, r)`, `np.argmax`, `plt.hist`

Think about. A cell in a dense tumour has its 10 nearest neighbours much closer than a cell in sparse stroma. Is this a feature or a problem? When would you prefer radius over k-NN?

Exercise 1.3: Compute Neighbourhood Composition Bags (25 min)

For each cell, summarise its neighbourhood as a vector of **cell-type fractions**. End result: a NumPy array of shape $(N, 9)$.

Tasks

1. Implement a **loop version**: for each cell, iterate over its k-NN neighbours, count cell types, normalise to fractions. Check a few individual rows.
2. Rewrite using **vectorised NumPy operations** (hint: one-hot encoding + array indexing). Verify both give the same result and compare running times.
3. Compute the **9 x 9 co-occurrence matrix**: for each (focal type, neighbour type) pair, what is the mean fraction of that neighbour type? Display as an annotated heatmap.
4. Interpret the heatmap:
 - Which cell types have the strongest self-similar neighbourhoods (high diagonal)?
 - Which pairs of cell types are most often found near each other?
 - Are any pairings surprising?

5. **Statistical significance of co-enrichment**

Start: For each cell type t , compute an enrichment ratio: the observed mean fraction of t among neighbours of t -cells divided by the global fraction of t in the tissue. Which cell types are most strongly self-enriched? Which are depleted (ratio < 1)?

Bonus: Build a permutation null: randomly shuffle all cell-type labels (positions fixed), recompute the full co-occurrence matrix, repeat 200 times. For each entry (i, j) compute a z-score: $(\text{observed} - \text{null_mean}) / \text{null_std}$. Display as a z-score heatmap. Which pairs survive $|z| > 3$ after Bonferroni correction for $9 \times 9 = 81$ tests?

Hint (relevant functions): `np.eye` (one-hot encoding), fancy indexing `array[index_array]`, `ndarray.mean(axis=1)`, `sns.heatmap, %timeit`

Think about. The diagonal is partly driven by global cell-type abundance; a common type will appear in many neighbourhoods by chance alone. How would you test whether the co-occurrence is more than expected by chance?

Session 1 check-in. Before continuing, make sure:

- You have a **bags** array of shape $(N, 9)$.
- Each row sums approximately to 1.
- The co-occurrence heatmap is non-uniform.

If any of these are not true, stop and debug now; later exercises build directly on this.

Session 2: Spatial Niches, Cohort Fingerprint and Niche Adjacency (90 min)

Goal. Define discrete spatial niches, determine which are shared across all 15 patients (the cohort fingerprint), and characterise which niches tend to be spatially adjacent to each other.

Exercise 2.1: Cluster Bags into Spatial Niches (20 min)

Tasks

1. L1-normalise the bag matrix so rows sum exactly to 1.
2. Run k-means for $k = 2, 3, \dots, 12$. Plot the **inertia (elbow plot)**. Do not rely on the elbow alone; think about what niches you expect in a lung tumour (tumour core, immune-excluded stroma, infiltrated border, lymphoid aggregates, ...).
3. Fit a final model with your chosen k . Display the **cluster centroids as a heatmap** (niches x cell types). Assign a short biological name to each niche.
4. Make a **UMAP projection** of a 50,000-cell subsample, coloured first by niche label, then by cell type. Does the UMAP show clean niche separation?

Hint (relevant functions): `sklearn.preprocessing.normalize`, `sklearn.cluster.MinibatchKMeans` (use this, not `KMeans`; it is much faster), `km.inertia_`, `km.cluster_centers_`, `umap.UMAP(metric="cosine")`, `sns.heatmap`

Think about. The number of niches k is a biological hypothesis, not a free parameter to tune. Could you defend your choice of k to a pathologist who knows this tissue?

Exercise 2.2: Map Niches Back onto the Tissue (15 min)

Tasks

1. Side-by-side scatter plot: tissue type on the left, niche labels on the right. Use a fixed colour per niche; keep it consistent across all patients.
2. For each niche, what fraction of cells are in **stroma** vs **tumor**? Display as a stacked bar chart.
3. Do niche boundaries align with the tissue-type boundary? Where do they diverge?
4. Try $k-1$ and $k+1$ niches. How does the spatial map change?

Hint (relevant functions): `plt.scatter`, `pd.crosstab(df["niche"], df["tissue.type"], normalize="index")`, `DataFrame.plot(kind="bar", stacked=True)`

Exercise 2.3: Cohort Fingerprint, Shared Niches Across Patients (25 min)

A niche that appears in only one patient could just as well be a technical quirk as a biological reality. Recurrence across patients is what makes a niche worth caring about.

Tasks

1. For each of the remaining 14 patients: compute their bag matrix and assign niche labels using the k-means model trained on patient 1 (transfer the clustering, do not refit).
2. For each patient, compute the **fraction of cells in each niche**. Display all 15 patients as a stacked bar chart.
3. Answer:
 - Which niches are present in *all* patients? Which are absent in some?
 - Which niche shows the greatest inter-patient variation in abundance?

- Is there a niche whose abundance seems to define individual patients?
4. Pick one niche and compute its **mean bag vector** for each patient separately. Use cosine similarity to measure how consistent the niche content is across patients.
 5. **Statistical significance of niche enrichment and inter-patient variability**
Start: For each niche, build a 2x2 contingency table: (in niche / not in niche) x (in tumour ROI / not in tumour ROI) pooled across all 15 patients. Use `scipy.stats.chi2_contingency` to test whether each niche is significantly enriched or depleted in the tumour compartment. Apply Bonferroni correction for the number of niches tested.
Bonus: Under Poisson sampling, the expected inter-patient variance in niche fraction for niche i is approximately $p_i (1 - p_i) / N$ where p_i is the cohort-mean fraction and N the patient cell count. Compute this lower bound for each niche and compare it to the observed variance across the 15 patients. Niches whose observed variance substantially exceeds the Poisson floor are *biologically variable*, not just sampling noise. Which niches are most biologically variable?

Hint (relevant functions): `km.predict(X_new)`, `scipy.stats.chi2_contingency`, `np.var(axis=0)`, `sklearn.preprocessing.normalize`, cosine similarity: `np.dot(a, b) / (np.linalg.norm(a) * np.linalg.norm(b))`, `pd.DataFrame`, `DataFrame.plot(kind="bar", stacked=True)`

Think about. If the same niche label has a very different bag composition in patient A vs patient B, what does that say about the robustness of the clustering?

Exercise 2.4: Niche Adjacency, What Neighbours What? (25 min)

Knowing which niches exist is only half the story. Where they sit relative to each other (what borders what) is the kind of question a pathologist reasons about qualitatively every time they look at a slide. We want a number for it.

Tasks

1. For each cell you know its niche and the niches of its k-NN neighbours. Build a **niche x niche adjacency matrix**: entry (i, j) is the mean fraction of niche- j cells found among the neighbours of niche- i cells.
2. Display the adjacency matrix as a heatmap.
3. Interpret the matrix:
 - Which niche pairs are most frequently adjacent?
 - Which pairs almost never neighbour each other (spatial exclusion)?
 - Does the tumour niche preferentially border immune niches or stroma niches?
4. Compute the same adjacency matrix for at least 4 additional patients. Are the adjacency patterns consistent or patient-specific? Summarise inter-patient variability (e.g. a heatmap of standard deviations, or box plots of individual adjacency entries across patients).
5. **Permutation test for spatial co-location**
Start: Pick the niche pair with the highest observed adjacency entry. Build a null distribution: randomly permute all niche labels (positions fixed), recompute that single adjacency entry, repeat 200 times. Plot the null as a histogram, mark the observed value, and compute a one-sided empirical p-value (fraction of permutations

$\geq$ observed).

Bonus: Repeat the permutation test for every entry of the $k \times k$ adjacency matrix. Apply Benjamini-Hochberg correction across all k^2 entries. Report which niche pairs show significant spatial **co-location** (adjacency higher than null) and which show significant spatial **exclusion** (adjacency lower than null). Do the same significant pairs appear across multiple patients, or are they patient-specific?

Hint (relevant functions): `knn_indices` (already built), `np.eye` + fancy indexing to look up neighbour niches, `pd.crosstab` or manual accumulation with `np.zeros`, `sns.heatmap`, `np.std(axis=0)`

Think about. The adjacency matrix is a spatial transition probability between niches. If you drew an arrow from each niche to its most frequent neighbour, what graph would emerge? Does this graph differ between patients?

Discovery Extension (Optional / Session 5)

Sessions 1 and 2 were about building the machinery. This section is about using it to say something about the biology. These exercises are optional, but if you are curious where this kind of analysis actually leads, this is worth trying. Each exercise has a **Start** task and a **Bonus** that is genuinely open-ended.

Exercise D1: How Deep Do Immune Cells Penetrate? (Spatial Infiltration Profiles)

An *immune-excluded* tumour has T cells piling up at the tumour edge but not getting in. An *inflamed* tumour has them distributed throughout the core. Pathologists reason about this distinction every time they look at a slide; we just want to make it quantitative and reproducible across all 15 patients.

Background. For each cell, define its **border distance** as the distance to the nearest cell of the *opposite* tissue type (tumour cell nearest to stroma, or stroma cell nearest to tumour). Positive values = deeper inside the tissue compartment.

Tasks

1. Build a `cKDTree` on stroma cells and query every tumour cell, and vice versa. Call the result `border_dist`. Add it as a new column to the dataframe.
2. Plot the **border distance distribution** separately for each cell type, using kernel density estimation or overlaid histograms. Focus on cells inside the tumour ROI. Where are T cells concentrated relative to the border?
3. Compute a **spatial infiltration profile**: bin `border_dist` into 50 μm bins; for each bin compute the fraction of T cells among all cells in that bin. Plot fraction vs. distance for one patient.
4. **Start:** Define a per-patient **T cell infiltration score** as the median `border_dist` of T cells that are *inside* the tumour ROI. A high positive score means T cells penetrate deep into the tumour core; a score near zero means they are trapped at the periphery. Compute this for all 15 patients and rank them.

Bonus: Plot the infiltration profiles (fraction vs. distance) for all 15 patients as a grid of line plots. Do patients fall into visually distinct classes (infiltrated / excluded / mixed)? Fit a simple exponential decay $f(d) = A * \exp(-d / \lambda)$ to each profile and use the decay length λ as a second, continuous infiltration score. Compare it to the median-based score from Start.

Hint (relevant functions): `ckdTree.query(coords, k=1)` (returns distances to nearest neighbour), `np.digitize`, `scipy.stats.gaussian_kde`, `scipy.optimize.curve_fit`

Think about. If you computed infiltration score for all 15 patients, do the extremes (most excluded vs most infiltrated) differ in any spatial niche fractions you found in Session 2? What might that mean for how the immune system “gets stuck”?

Exercise D2: Moran’s I, a Formal Test of Spatial Clustering

You have already seen visually that T cells cluster. Moran’s I gives a single number that quantifies *how much* any variable is spatially autocorrelated – and lets you test whether the clustering is more than chance.

For a cell-type indicator I_i (1 if cell i is of the target type, 0 otherwise), Moran’s I is:

$$\mathcal{I} = \frac{N}{W} \cdot \frac{\sum_{ij} w_{ij} (I_i - \bar{I})(I_j - \bar{I})}{\sum_i (I_i - \bar{I})^2}$$

where $w_{ij} = 1$ if j is among the k -NN of i , and $W = \sum_{ij} w_{ij}$.

Tasks

1. Implement `morans_i(indicator, knn_indices)` that returns a scalar. Test on a synthetic case: a perfectly clustered indicator should give I near +1; a checkerboard pattern near -1; a random shuffle near 0.
2. **Start:** Compute Moran’s I for each of the 9 cell types on one patient (work on a 50,000-cell subsample to keep it fast). Build a permutation null (200 random shuffles of the indicator) to get an empirical p-value. Display the results as a bar chart of I values with error bars from the null. Which cell types cluster most strongly? Are any dispersed?
Bonus: Compute Moran’s I for each cell type across all 15 patients (subsample each). Display as a boxplot: patients on x-axis, I on y-axis, one panel per cell type. Is the degree of T cell clustering consistent across patients, or does it vary? Does a patient with high T cell I also tend to have high T cell infiltration score from D1?

Hint (relevant functions): `knn_indices` (already built), vectorised inner product as `np.einsum` or sparse matrix (`scipy.sparse.csr_matrix`), `np.random.permutation`

Think about. Moran’s I is a *global* statistic: it summarises the whole slide in one number. How would you detect a situation where T cells cluster in *one region* of the slide but are dispersed everywhere else?

Exercise D3: Patient Stratification from Spatial Features

Everything we built (niches, adjacency, infiltration, autocorrelation) can be collapsed into a single row of numbers per patient. Fifteen patients, fifteen rows. Do those rows cluster? If they do, you have found patient subtypes defined entirely by spatial microenvironment, with no pathologist labels involved.

Tasks

1. Assemble a **spatial feature matrix** of shape (15 patients x F features):
 - Niche fractions from Exercise 2.3 (k features).
 - Diagonal of the niche adjacency matrix from Exercise 2.4 (k features: how often does each niche border itself?).
 - T cell infiltration score from Exercise D1 (1 feature).
 - Moran's I for T cells from Exercise D2 (1 feature).
2. Z-score each feature across patients. Run **PCA** on the standardised matrix. Plot all 15 patients in PC1-PC2 space (label each point with the patient ID). What fraction of variance does PC1 explain?
3. Inspect the PC1 loadings. Which spatial features drive the main axis of patient variation? Is it dominated by niche composition, adjacency, or infiltration?
4. **Start:** Cluster the 15 patients into 2 or 3 groups using k-means on the first 3 PCs. For each group, compute the group-mean spatial map (average niche fraction at each tissue type). Does each group have a recognisable spatial phenotype? Assign a short biological label (e.g. "immune desert", "inflamed", "mixed").
Bonus: The cohort likely varies in histological subtype or other clinical properties (check the README or file naming). Test whether the patient clusters from Start align with any available annotation using Fisher's exact test. If no annotation is available, design a statistical test you *would* run if you had access to 5-year overall survival data – write it as a pre-registration paragraph (one paragraph, not code).

Hint (relevant functions): `sklearn.preprocessing.StandardScaler`, `sklearn.decomposition.PCA`, `pca.explained_variance_ratio_`, `pca.components_`, `scipy.stats.fisher_exact`

Think about. This entire feature space was built from nothing but x, y coordinates and cell-type labels, with no pixel intensities and no pathologist annotations. If patients separate here, what does that tell us?

Final Discussion

1. **Cohort fingerprint.** Which niches are universal and which are variable across patients? What might drive these differences?
 2. **Niche adjacency.** Is there a consistent spatial grammar, a preferred ordering of niches from tumour core outward? What would this suggest about tumour microenvironment organisation?
 3. **Patches vs. neighbourhoods.** Name one biological question that last week's patch approach answers better, and one that this week's neighbourhood approach answers better.
 4. **Next steps.** If you had access to patient survival data, how would you test whether niche abundance or niche adjacency predicts clinical outcome?
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Quick Reference

Task	Key function / approach
Spatial index	<code>scipy.spatial.cKDTree(coords)</code>
k-NN query	<code>tree.query(coords, k+1)</code> (exclude self with <code>[:, 1:]</code>)
Radius query	<code>tree.query_ball_point(coords, r)</code> (remove self from each list)
One-hot encoding	<code>np.eye(n_types)[ct_array]</code>
Vectorised bags	<code>one_hot[knn_indices].mean(axis=1)</code>
Clustering (large N)	<code>sklearn.cluster.MinibatchKMeans</code>
L1 normalisation	<code>sklearn.preprocessing.normalize(X, norm="l1")</code>
UMAP	<code>umap.UMAP(metric="cosine").fit_transform(X)</code>
Transfer clustering	<code>km.predict(X_new)</code>
Cross-tabulation	<code>pd.crosstab(df["niche"], df["tissue.type"], normalize="index")</code>
Cosine similarity	<code>np.dot(a, b) / (np.linalg.norm(a) * np.linalg.norm(b))</code>
Enrichment ratio	observed fraction / global fraction
Permutation null	shuffle labels with <code>np.random.permutation</code> , loop 200x
Chi-squared test	<code>scipy.stats.chi2_contingency(table)</code>
Bonferroni	<code>p_corrected = p * n_tests</code>
Benjamini-Hochberg	<code>statsmodels.stats.multitest.multipletests(pvals, method="fdr_bh")</code>
Poisson variance floor	$p * (1 - p) / N$ where p = mean fraction, N = cell count
Distance to boundary	<code>cKDTree(stroma_coords).query(tumor_coords, k=1)[0]</code>
Moran's I weights	<code>scipy.sparse.csr_matrix</code> from knn edges
PCA on patients	<code>sklearn.decomposition.PCA</code> , inspect <code>.components_</code> for loadings
Standardise features	<code>sklearn.preprocessing.StandardScaler</code>
Curve fitting	<code>scipy.optimize.curve_fit(f, x, y, p0=[...])</code>
Fisher's exact test	<code>scipy.stats.fisher_exact([[a,b],[c,d]])</code>

If something does not work as expected, that is usually the most interesting part.